

Changing states

When solids get hot enough, they melt and become liquids. When liquids get cold enough, they freeze and become solid. This is called changing states and it happens to all kinds of substances.

Liquid metal

Every substance melts and boils at a particular temperature (its melting and boiling points). Most metals are solid at everyday temperatures because they have a high melting point. But mercury has such a low melting point that it is liquid even at room temperature.

Condensation

As water vapour in the air is cooled, it changes into liquid water. This is called condensation. You can see it on the outside of a cold bottle.

When water vapour in the air touches a cold bottle, it condenses into tiny drops of liquid.

Changing states of water

Water exists as a solid, liquid, or gas. You can find all three forms of water in your home. They are ice, water, and steam (water vapour).



Ice is solid water. It forms when liquid is cooled until it freezes. Each piece of ice has a definite shape.



When ice is warmed, it melts and becomes liquid and takes on the shape of the container holding it.



As water is heated, bubbles of steam form. They rise to the surface and burst, so steam escapes into the air.



Rivers of iron

Iron must be heated in a furnace to make it melt. Molten iron is so hot it glows white. It is poured into a mould and left to harden to make solid iron objects.